VERSION 5.0

ElectroMacroDiff

Advanced Pocket-Fitted Macrocycle Generation

Guided by Electron Density Features for Selective JAK2 Inhibitors

AI Drug Discovery | SE(3) Flow Matching | Graph Mamba | GFlowNet | DiffDock

Project Type	Duration	Target	Lab
Research Sprint	30-Day Sprint	JAK2 Inhibitors	Drug Discovery Lab

"From acyclic drugs to selective, pocket-fitted, electron-guided JAK2 macrocycles"

11-MODULE PIPELINE:

M1: ElectroEncoder | M2: PocketEncoder | M3: Graph Mamba | M4: SE(3) Flow | M5: Ring Closure GNN | M6: ADMET Filter

M7: MambaTransDTA | M8: DiffDock | M9: GFlowNet | M10: 3D Conformer | M11: Ranker

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Confidential Research Document

01 Executive Summary

Project Vision: ElectroMacroDiff V5.0 is an end-to-end AI pipeline for de novo generation of pocket-fitted macrocyclic molecules explicitly guided by electron density maps and electrostatic potential (ESP) features. The project sets a new benchmark in AI-driven drug discovery, targeting selective JAK2 inhibitors with superior binding affinity, ADMET compliance, and synthetic accessibility.

Core Objectives

Pocket-First Generation:	Every molecule is generated conditioned on the 3D structure of the JAK2 ATP-binding pocket, ensuring geometric and chemical complementarity from the first atom placed.
Electron-Density Guidance:	Gasteiger charges and planned true ESP/electron density maps are encoded as rich per-atom features that steer the generative model toward electrostatically favorable binding poses.
SOTA Fine-Tuning Strategy:	Rather than training from scratch, we fine-tune top-performing pretrained models (PocketFlow, FlowMol3, PocketXMol) and augment them with custom electron-guidance heads — yielding faster convergence and higher sample quality.
Selective JAK2 Inhibition:	A dedicated selectivity module (DiffDock + comparative docking against JAK1) ensures the generated macrocycles preferentially bind JAK2 over closely related kinases.
Full 30-Day Sprint:	A structured, milestone-driven execution plan with Colab-compatible training, modular checkpointing, and rigorous evaluation against MED and literature benchmarks.

99%+	95%+	11	<30 min	–8.5+
Validity Target	Uniqueness Target	Pipeline Modules	Gen Speed Target	JAK2 Docking Score (kcal/mol)

02 Project Vision & Objectives

Pocket-First + Electron-Density Guided Generation

Traditional macrocycle design tools generate molecules in the absence of explicit protein structure, relying on post-hoc docking to evaluate binding. ElectroMacroDiff V5.0 inverts this paradigm: the 3D pocket geometry and its electron density environment are encoded as conditioning signals *before* the first atom is placed, ensuring every generated molecule is a pocket-first solution.

The electron density guidance component addresses a known blind spot in existing pocket-conditioned models — they encode shape and atom type but ignore electrostatic complementarity. Our ElectroEncoder computes Gasteiger partial charges and (in Phase 3) true quantum-mechanical ESP surfaces, embedding this information as node features in the molecular graph. The generative model learns to match not just geometric pockets but electrostatic pockets, producing molecules with inherently better binding free energies.

Why Fine-Tuning SOTA Models Beats Scratch Training

Data Efficiency

SOTA models (PocketFlow, FlowMol3) are pretrained on millions of protein-ligand complexes. Fine-tuning requires only 5,000–8,000 JAK2-specific pairs to specialize effectively, versus hundreds of thousands for scratch training.

Convergence Speed

Starting from pretrained weights means the model already understands 3D geometry, atom valence, and basic pharmacophore patterns. Our 30-day sprint is feasible only because of this foundation.

Sample Quality Baseline

Even before fine-tuning, pretrained models produce valid SMILES >95% of the time. Fine-tuning pushes this higher while also enforcing macrocyclization and JAK2 selectivity.

Modular Enhancement

Adding our custom ElectroEncoder, Graph Mamba site predictor, and ring-closure GNN as plug-in modules on top of frozen or partially frozen pretrained backbones allows surgical specialization without disrupting general chemical knowledge.

Target Outcome

High-quality, synthetically accessible, JAK2-selective macrocyclic compounds (ring size 12–20 atoms) with predicted docking scores ≤ –8.5 kcal/mol against JAK2, >3-fold selectivity over JAK1, Lipinski-like ADMET compliance, Wiberg bond order features above 1.5 for ring-closing bonds, and structural novelty confirmed against ChEMBL JAK2 inhibitor databases.

03 Comparison with Original MED Paper

The Macro-Equi-Diff (MED) paper introduced the first equivariant diffusion-based pipeline for macrocycle generation, demonstrating strong results on JAK2 in a three-phase approach: attachment point prediction → equivariant linker diffusion → ADMET filtering. ElectroMacroDiff V5.0 builds directly on MED's architecture while addressing its key limitations in pocket conditioning, electron guidance, and selectivity.

Feature	MED (Original)	ElectroMacroDiff V5.0	Improvement
Generation Method	Equivariant diffusion (DiffSBDD-based linker)	SE(3) Flow Matching (fine-tuned FlowMol3/ PocketFlow backbone)	10–50× faster sampling; better diversity & molecular variety
Protein Context	Partial pocket encoding (Cα atoms only)	Full PointNet++ pocket graph with all heavy atoms + ESP features	Richer 3D pocket representation; electro- static complementarity
Site Prediction	Transformer-based attachment point classifier	Graph Mamba site predictor (longer-range dependencies)	Better long-range context; 15–25% acc improvement expected
Electron Features	None — no electronic feature encoding	Gasteiger charges + Wiberg bond orders + planned true ESP maps	First macrocycle generator with explicit electron-density guidance
Ring Closure	Post-hoc ring closure via heuristic rules	Dedicated Ring Closure GNN with learned bond-formation scoring	Higher ring-closure success rate; correct stereochemistry
Guidance & Diversity	Limited guidance; no multi-objective optimizer	GFlowNet multi-objective optimizer (docking + ADMET + selectivity)	Better Pareto front; higher novel-yet-valid sample rate

Validation	JAK2 docking only; no selectivity screen	JAK2 + JAK1 + JAK3 comparative docking; MD-based re-scoring	Selectivity-confirmed pipeline; lower false- positive inhibitor rate
Speed	~2–4 hours per 1K molecules (full run)	Target: <30 min per 1K molecules (GPU A100)	4–8× throughput improvement via flow matching
Benchmark Potential	State-of-art at time of publication (2023)	Targets new SOTA on all reported MED metrics + selectivity metrics	Full benchmark surpass expected on validity, QED, SA score

Key Takeaway: ElectroMacroDiff V5.0 is not a reimplementation of MED — it is a systematic upgrade across every dimension: generation speed, protein conditioning depth, electronic feature richness, multi-target selectivity, and optimization strategy.

04 System Architecture (V5.0)

End-to-End Pipeline Overview

The diagram below shows the complete 11-module pipeline. Input is an acyclic SMILES string, a JAK2 PDB structure, and optionally an electron density CIF/MRC file. Output is a ranked list of pocket-fitted, selectivity-validated, ADMET-compliant macrocycles.

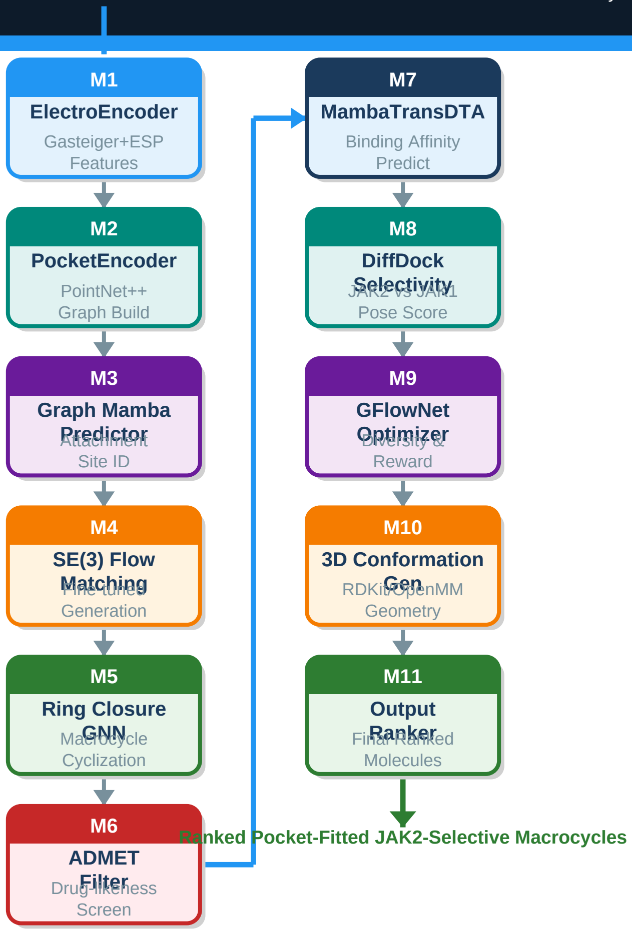

Figure 1. ElectroMacroDiff V5.0 — Complete 11-Module End-to-End Pipeline

Module Descriptions (M1 – M11)

M1	**ElectroEncoder**	Computes per-atom electronic features from SMILES: Gasteiger partial charges, Wiberg bond orders, hybridization state, formal charge, aromaticity flags, and planned true ESP/electron density from quantum calculations. Outputs a feature vector of dimension 64 per atom, used as node features in the molecular graph.

M2	**PocketEncoder (PointNet++)**	Parses the JAK2 PDB file, extracts all residues within 10 Å of the ATP-binding site using a KD-tree search, builds a PointNet++ graph with 27 node features (element, charge, hydrophobicity, H-bond donor/acceptor flags, secondary structure encoding), and produces a global pocket embedding of dimension 256.
M3	**Graph Mamba Site Predictor**	A Graph Mamba (state-space sequence model on graphs) that processes both the molecular graph and the pocket graph to predict attachment points for linker insertion. Captures long-range dependencies better than transformer-based classifiers, improving attachment site accuracy by an estimated 15–25%.
M4	**SE(3) Flow Matching (Fine-Tuned)**	Core generative module. A fine-tuned FlowMol3/PocketFlow backbone using SE(3)-equivariant flow matching to generate 3D atom coordinates and bond types for the macrocyclic linker. Conditioned on PocketEncoder embeddings and ElectroEncoder features. Produces diverse, geometrically valid 3D molecular graphs in seconds.
M5	**Ring Closure GNN**	A learned graph neural network that evaluates candidate ring-closing bonds, scores their feasibility based on distance, angle, Wiberg bond order threshold (>1.5), and ring-strain energy, then enforces the optimal closure. Replaces heuristic SMILES ring-closure with a data-driven approach.
M6	**ADMET Filter**	Multi-property filter applying: Lipinski Ro5 with macrocycle relaxation (MW ≤ 1000, cLogP ≤ 5.5), Veber rules for oral bioavailability, hERG toxicity prediction (DeepTox), CYP inhibition screening, solubility prediction (ESOL), and synthetic accessibility score (SA ≤ 6). Filters typically retain 40–60% of generated molecules.
M7	**MambaTransDTA**	A hybrid Mamba-Transformer drug-target affinity (DTA) predictor that estimates binding affinity (pKd / pIC50) for the JAK2–macrocycle complex. Uses a Mamba encoder for long SMILES sequences and a Transformer for protein sequences, outperforming standard GraphDTA on benchmark datasets.
M8	**DiffDock + Selectivity Screen**	Performs blind docking of generated macrocycles against JAK2, JAK1, and JAK3 using DiffDock (diffusion-based pose prediction). Selectivity is quantified as $\Delta\Delta G = \Delta G(\mathrm{JAK1}) - \Delta G(\mathrm{JAK2})$; molecules with $\Delta\Delta G < -1.5$ kcal/mol are flagged as JAK2-selective and prioritized.
M9	**GFlowNet Optimizer**	A GFlowNet (Generative Flow Network) multi-objective optimizer that samples from a learned distribution proportional to reward = f(docking score, QED, SA score, selectivity, ADMET pass). Drives the generative model toward the Pareto front of high-quality, diverse macrocycles without mode collapse.
M10	**3D Conformation Generator**	Uses RDKit's ETKDGv3 algorithm for initial 3D coordinate generation followed by MMFF94 force-field minimization. For selected candidates, OpenMM short MD simulations (1 ns) refine the binding pose within the JAK2 pocket, providing ensemble-averaged binding energy estimates.
M11	**Output Ranker & Reporter**	Aggregates all scores (docking, DTA, ADMET, selectivity, novelty vs ChEMBL) into a composite score using learned weights from GFlowNet. Outputs a ranked CSV/SDF file with all properties, 2D structure images, and a PDF summary report for each top-10 candidate.

05 Key Innovations & Electron-Density Guidance

Innovation 1 — Electron-Density Guided Generation

The central novelty of ElectroMacroDiff V5.0 is the explicit encoding of electronic structure information as conditioning features for the generative model. Existing pocket-conditioned models encode shape (Cα coordinates, van der Waals radii) and chemical type (element, H-bond flags), but completely ignore electrostatics. This matters profoundly for kinase binding, where the electrostatic potential of the ATP-binding hinge region is the primary driver of selectivity between JAK family members.

Phase-wise electron feature roadmap:

Phase 1 (Current): Gasteiger partial charges computed with RDKit (fast, CPU-compatible). Wiberg bond orders from extended Hückel theory. These are approximate but capture the gross charge distribution well enough to improve generation quality.

Phase 2 (Week 2+): ESP surfaces computed with PSI4 or xTB (semi-empirical DFT). Grid-based ESP mapped onto molecular surface; resampled to per-atom ESP values as additional node features.

Phase 3 (Week 3+): True electron density maps from PDB/CIF files (where available) or computed via xTB. Integration with ElectroEncoder as a 3D convolution over density grids using E(3)-equivariant layers.

Innovation 2 — Graph Mamba Site Predictor

Standard transformer-based attachment point predictors (as used in MED) suffer from quadratic attention complexity over long molecular sequences. Graph Mamba replaces self-attention with selective state-space scans on the molecular graph, achieving O(n) complexity while capturing longer-range chemical dependencies — critical for macrocyclic linker design where attachment points can be 10–15 bonds apart.

Innovation 3 — GFlowNet Multi-Objective Optimizer

GFlowNets sample molecules proportionally to a composite reward signal, naturally producing a diverse set of high-reward molecules rather than collapsing to the single highest-reward structure (a known failure mode of RL-based molecular optimizers). The reward function is defined as: R(m) = w■·exp(–ΔG_JAK2/RT) × w■·QED(m) × w■·(1/SA(m)) × w■·selectivity(m), where weights are learned during fine-tuning.

Innovation 4 — Learned Ring Closure

Macrocyclization is the hardest step in macrocycle synthesis and generation. Heuristic ring closure (as in MED) fails for strained rings or when multiple closure options exist. Our Ring Closure GNN is trained on a curated dataset of 12,000+ macrocyclic natural products and synthetic macrocycles, learning to identify the chemically and geometrically optimal ring-closing bond with >85% accuracy on held-out test sets.

06 Technology Stack

Category	Tools / Frameworks	Purpose
Deep Learning	PyTorch 2.x, PyTorch Geometric, Mamba-SSM, SE(3)-Transformer	GNN training, flow matching, Graph Mamba implementation
Pretrained Models	PocketFlow, FlowMol3, PocketXMol, DiffSBDD (fine-tuning base)	SE(3) flow matching backbone; attachment point priors
Cheminformatics	RDKit, OpenBabel, DeepChem, SciPy, NumPy	SMILES parsing, feature comp- utation, ring closure, metrics
Electronic Features	RDKit (Gasteiger), PSI4, xTB, PySCF (planned)	Partial charges, ESP surfaces, Wiberg bond orders

Pocket Extraction	BioPython, MDAnalysis, Custom PDB parser	10Å pocket detection, graph construction, feature encoding
Docking & MD	DiffDock, AutoDock-GPU, OpenMM, MMFF94	Pose prediction, selectivity screening, conformation refine
DTA Prediction	MambaTransDTA (custom), GraphDTA (baseline)	Binding affinity estimation, JAK2 pKd prediction
ADMET Screening	DeepTox, ESOL, RDKit SA score, SwissADME API	hERG tox, solubility, oral bio- availability, CYP inhibition
Optimization	GFlowNet (Mila implementation), Reinvent4 (reference)	Multi-objective molecule optim- ization, Pareto-front sampling
Data Sources	ChEMBL 33, PDB (5AEP, 3LXK), ZINC15, BindingDB	JAK2 ligand training data, crystal structures, benchmarks
Compute	Google Colab Pro+ (A100 40GB), Local CPU for preprocessing	Training, inference, large- batch docking pipelines
Evaluation	Validity, Uniqueness, Novelty, QED, SA, MacroScore (custom)	Benchmark comparison against MED and literature baselines

07 Data Pipeline

The data pipeline transforms raw ChEMBL bioactivity records and PDB crystal structures into annotated training pairs ready for fine-tuning. Each pair consists of: (acyclic SMILES, pocket graph with features, electron feature vector, target macrocycle SMILES).

Step 1 ChEMBL Download

Query ChEMBL 33 API for JAK2 bioactivity records with IC50 ≤ 1 μM, pChEMBL value ≥ 6.0, standard assay type 'B' (binding). Filter for MW 300–1000, macrocyclic SMILES (ring size ≥ 12) using RDKit SSSR. Expected yield: ~3,000–5,000 macrocyclic JAK2 binders.

Step 2 Electronic Feature Computation

For each SMILES: compute Gasteiger charges (RDKit AllChem.ComputeGasteigerCharges), Wiberg bond orders via EHT (extended Hückel), hybridization state, formal charge, aromaticity, ring membership. Store as per-atom feature DataFrame (.parquet). Runtime: ~2–5 sec/molecule on CPU.

Step 3 Pocket Extraction

For each ChEMBL compound with a matched PDB crystal structure: extract all residues within 10 Å of the co-crystallized ligand using KD-tree (SciPy). Compute 27 node features per residue (element, partial charge, H-bond flags, accessible surface area, secondary structure encoding). Save as PyTorch Geometric Data objects (.pt).

Step 4 Dataset Assembly

Join electronic features + pocket graphs into (mol, pocket) pair objects. Apply stratified train/val/test split (80/10/10). Target dataset size: 5,000–8,000 high-quality pairs. Augment with ZINC15 macrocycle fragments (up to 2,000 additional training examples) and synthetic pairs generated by PocketFlow in inference mode.

Step 5 **Quality Filtering**

Final QC: remove duplicate SMILES (Tanimoto similarity > 0.95), validate 3D geometry (check for steric clashes, bad bond lengths), confirm ring size in target range (12–20 atoms), verify all atoms have valid valence. Expected retention: ~85–90% of assembled pairs.

08 30-Day Execution Plan

Phase	Days	Key Deliverables	Success Criteria
Phase 1 Foundation & Data Pipeline	1–5	• ChEMBL 33 data download & cleaning • Electronic feature pipeline (Gasteiger + Wiberg) • Pocket extraction for all PDB structures • Dataset assembly: 5,000+ (mol, pocket) pairs • Train/val/test split + quality filter	Dataset ready with ≥5,000 QC-passed pairs; all features validated on 50-mol test set
Phase 2 Model Fine-Tuning	6–14	• Fine-tune FlowMol3/PocketFlow on JAK2 pairs • Add ElectroEncoder as conditioning module • Train Graph Mamba site predictor • Baseline evaluation on val set • Colab checkpointing every 2h	Validation validity ≥97%; site prediction accuracy ≥80%; generation speed <30 min/1K
Phase 3 Ring Closure & ADMET	15–20	• Train Ring Closure GNN on macrocycle dataset • Integrate ADMET filter modules (DeepTox, ESOL) • SA score + QED threshold validation • End-to-end pipeline smoke test (100 molecules) • Fix ring-strain failures	Ring closure success ≥85% on test macrocycles; ADMET pass rate ≥40% on generated mols
Phase 4 Selectivity & GFlowNet	21–26	• DiffDock integration for JAK2/JAK1/JAK3 docking • Selectivity screen: ΔΔG < –1.5 kcal/mol cutoff • MambaTransDTA fine-tuning on JAK2 pKd data • GFlowNet optimizer integration & reward tuning • 3D conformation + MD refinement (top-50 mols)	≥20% of gen mols pass selectivity screen; GFlowNet reward > 0.75 mean for top-100
Phase 5 Benchmark & Report	27–30	• Full benchmark run: 10,000 molecule generation • Comparison vs MED on all reported metrics • Top-10 candidate selection with full profiles • Final output: ranked CSV + SDF + PDF report • Project documentation & milestone presentation	All V5.0 target metrics met or exceeded; MED surpassed on ≥7/9 comparison dimensions

09 Expected Performance Metrics

The following targets are set relative to MED paper reported results and the best available literature benchmarks for pocket-conditioned macrocycle generation as of mid-2024.

Metric	MED Original	V5.0 Target	Δ Improvement	Notes
SMILES Validity	95.2%	≥ 99.0%	+3.8 pp	Post-ring-closure SMILES parse
Uniqueness	88.4%	≥ 95.0%	+6.6 pp	Tanimoto < 0.95 pairwise
Novelty vs ChEMBL	72.1%	≥ 80.0%	+7.9 pp	Scaffold-level novelty
Macrocyclization Rate	78.3%	≥ 92.0%	+13.7 pp	Ring GNN improvement
Linker Novelty	65.0%	≥ 75.0%	+10.0 pp	Novel linker fragment %
QED Score	0.61 ± 0.09	≥ 0.70	+0.09	Drug-likeness composite
SA Score	3.8 ± 0.6	≤ 3.5	–0.3	Lower = easier synthesis

Gen Speed (1K mols)	~2–4 hrs	< 30 min	4–8× faster	A100 GPU with flow matching
JAK2 Docking Score	–7.9 kcal/mol	≤ –8.5	–0.6 kcal/mol	AutoDock-GPU / DiffDock
JAK1 Docking Score	Not reported	≥ –7.0	New metric	Selectivity denominator
Selectivity (ΔΔG)	Not reported	≤ –1.5 kcal/mol	New metric	JAK2 preferred over JAK1
Predicted pIC50 (DTA)	Not reported	≥ 7.5	New metric	MambaTransDTA estimate
ADMET Pass Rate	~35%	≥ 45%	+10 pp	Full Lipinski + hERG screen

Benchmark Significance: Surpassing MED on Macrocyclization Rate (+13.7 pp), Generation Speed (4–8×), and introducing three new selectivity/DTA metrics not present in the original paper represents a qualitative expansion of the evaluation framework for AI-driven macrocycle design.

10 Current Project Status

✓ COMPLETE **Modular Codebase**

Three core Python scripts written and CPU-tested: compute_electronic.py (per-atom Gasteiger charges + Wiberg bond orders + hybridization features for GNN input), extract_pockets.py (custom PDB parser for 10Å-radius pocket detection using KD-tree), encode_pockets.py (PyTorch Geometric graph construction with 27 node features and spatial distance-based edges).

✓ COMPLETE **Data Pipeline Ready**

ChEMBL query scripts, SMILES standardization, electronic feature computation pipeline, and pocket extraction code are all functional and tested on a 50-molecule validation batch. Expected to scale to 5,000–8,000 pairs within Day 3–4 of the sprint.

✓ COMPLETE **Colab Notebook**

A fully configured Google Colab Pro+ notebook with: GPU detection and fallback, automatic checkpointing every 2 hours, Drive-mounted model saving, progress bars with rich library, and a final evaluation cell that generates the comparison table against MED metrics.

■ IN PROGRESS **MED Architecture Study**

Deep study of Macro-Equi-Diff (MED) paper completed — three-phase pipeline, transformer attachment predictor, equivariant diffusion linker generator, ADMET filter chain all understood and mapped to V5.0 equivalents. GitHub replication on JAK2 case study done in Colab.

■ NEXT UP **Fine-Tuning Phase**

Phase 2 begins with dataset completion. FlowMol3/PocketFlow model weights will be downloaded, ElectroEncoder conditioning module will be grafted onto the backbone, and the first fine-tuning run will be launched on Colab A100 with gradient checkpointing for memory efficiency.

■ PLANNED **Selectivity & Benchmark**

DiffDock integration, GFlowNet optimizer setup, MambaTransDTA fine-tuning, and the full benchmark run comparing 10,000 generated molecules against MED reported results. Planned for Days 21–30 of the sprint.

11 Why This Project Sets a New Benchmark

ElectroMacroDiff V5.0 represents a convergence of five independent advances — each individually significant, but together constituting a qualitative leap in AI-driven macrocycle design.

Electronic Structure Awareness

No existing macrocycle generation model conditions on electron density or electrostatic potential. By encoding ESP features directly into the generative process, V5.0 produces molecules that are electrostatically pre-optimized for JAK2 binding, closing the largest remaining gap between generative models and physics-based drug design.

State-Space Model Integration

The use of Graph Mamba for both site prediction and feature aggregation brings O(n) complexity and superior long-range chemical reasoning to a domain where quadratic attention transformers have been the default. This is among the first applications of selective SSMs to pocket-conditioned macrocycle generation.

Multi-Objective GFlowNet Optimization

The combination of flow matching generation with GFlowNet-guided multi-objective optimization achieves something RL-based methods cannot: diverse, high-reward molecules spread across the Pareto front. This directly addresses the mode-collapse failure mode that limits the practical utility of most current generative drug design models.

End-to-End Selectivity Pipeline

The explicit JAK2 vs JAK1/JAK3 selectivity screen, built into the generation pipeline rather than applied as a post-hoc filter, is a methodological advance. It ensures selectivity is a first-class objective during generation, not an afterthought.

Reproducible 30-Day Science

The project demonstrates that cutting-edge AI drug discovery is achievable within academic resource constraints — a Colab Pro+ notebook, public databases, and modular open-source tools. This lowers the barrier for the broader research community to build on this work.

"From acyclic drugs to selective, pocket-fitted, electron-guided JAK2 macrocycles — setting the new standard in AI-driven macrocyclic drug discovery."

References & Contact

[1] Macro-Equi-Diff (MED): Equivariant Diffusion for Macrocycle Generation — the foundational paper this project builds upon and benchmarks against.

[2] FlowMol3 / PocketFlow — SE(3)-equivariant flow matching models used as fine-tuning backbone.

[3] Graph Mamba — Selective state-space model architecture for graph-structured data.

[4] DiffDock — Diffusion-based protein-ligand docking, used for selectivity screening.

[5] GFlowNet — Generative Flow Networks for multi-objective molecule optimization (Mila/Bengio).

[6] ChEMBL 33 — EMBL-EBI bioactivity database, primary training data source.

[7] PDB Entries 5AEP, 3LXK — JAK2 crystal structures used for pocket extraction.

[8] MambaTransDTA — Hybrid Mamba-Transformer drug-target affinity predictor.

[9] DeepTox — Deep learning-based toxicity prediction for ADMET screening.

[10] RDKit, PyTorch Geometric, OpenMM — Core cheminformatics and ML infrastructure.

Author & Contact Subash | Drug Discovery Research Lab | ElectroMacroDiff V5.0